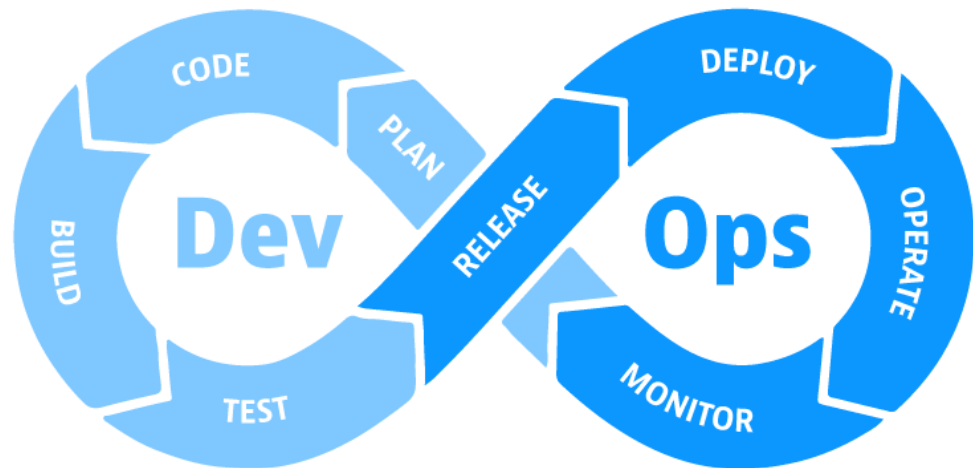
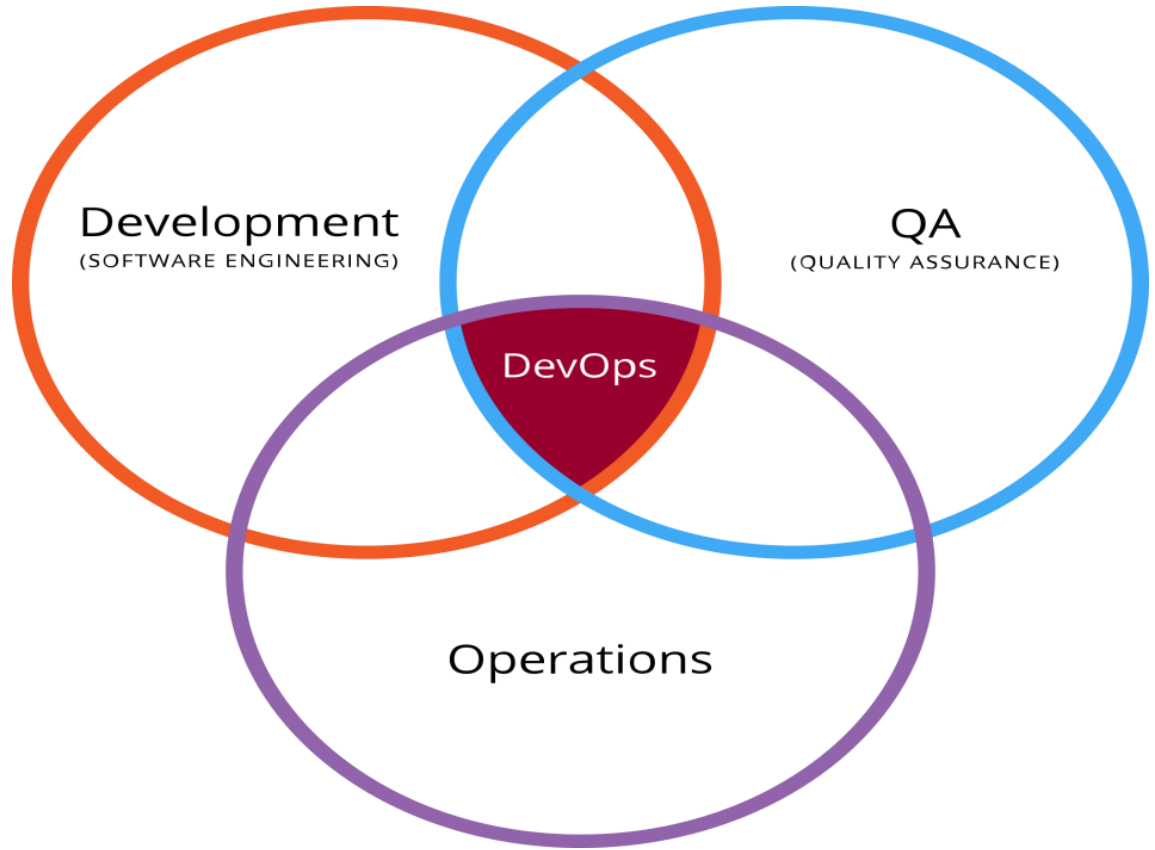


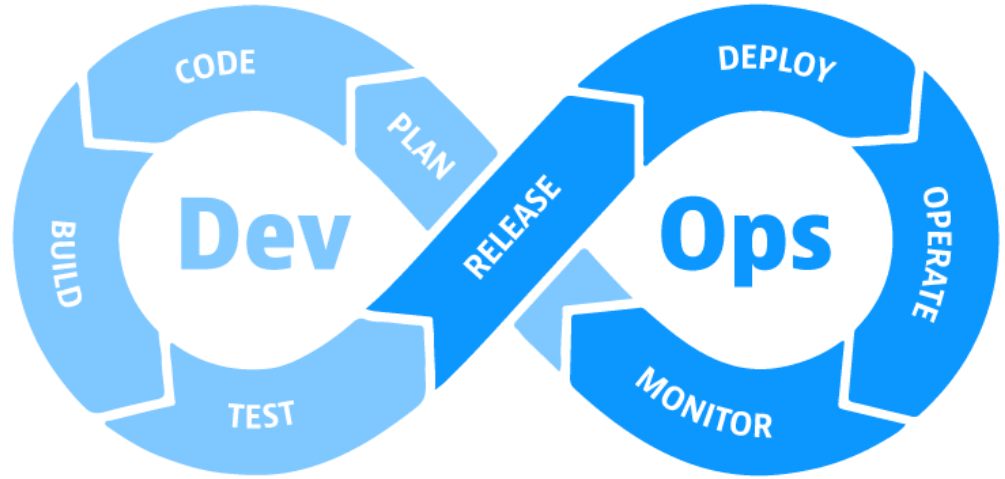


Introduction to DevOps



What is DevOps?

- DevOps (development and operations) is an enterprise software development phrase used to mean a type of agile relationship between development and IT operations.



What's and Why's of DevOps

- The goal of DevOps is to change and improve the relationship by advocating
- better communication and collaboration between the two business units.
- DevOps practices and procedures lead to smoothing out the typically 'bumpiest' aspects of software development and deployment.



What's and Why's of DevOps

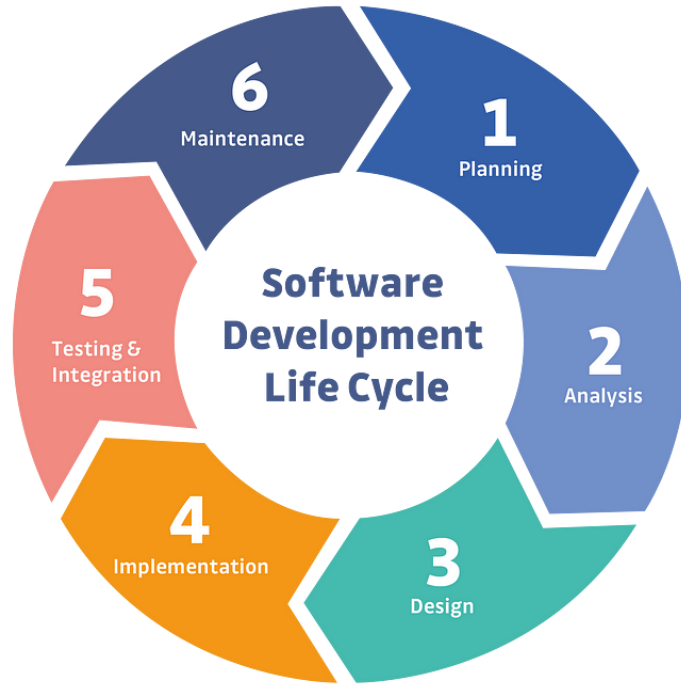
- By pulling infrastructure setup and awareness earlier in the development cycle, surprises from environmental differences are significantly reduced.
- By establishing consistent, reliable and repeatable automated deployments, human error and the need for 'rockstar firefighters' are reduced.



What's and Why's of DevOps

- In many ways, DevOps is about ensuring quality at all stages.





SDLC (Software Development Life Cycle):-

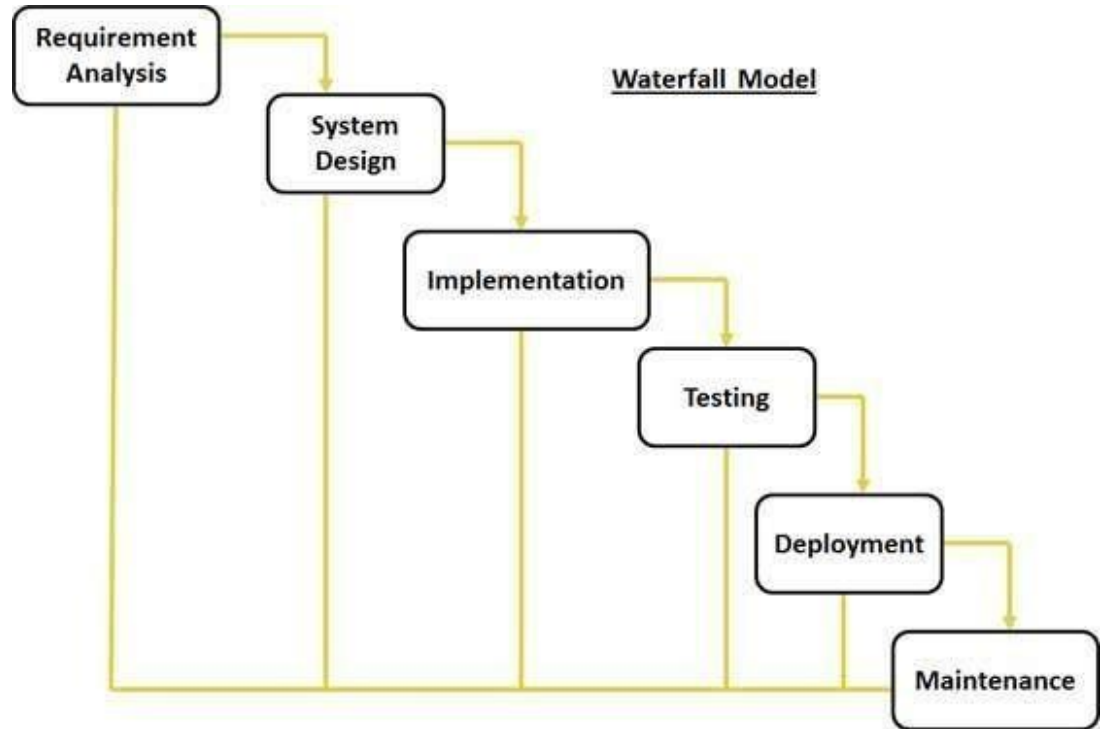
- Software Development Life Cycle is a process for designing, developing and testing enterprise applications to ensure high quality.

SDLC Models

- 
- Waterfall Model
 - Iterative Model
 - RAD Model
 - V Model
 - Prototype Model

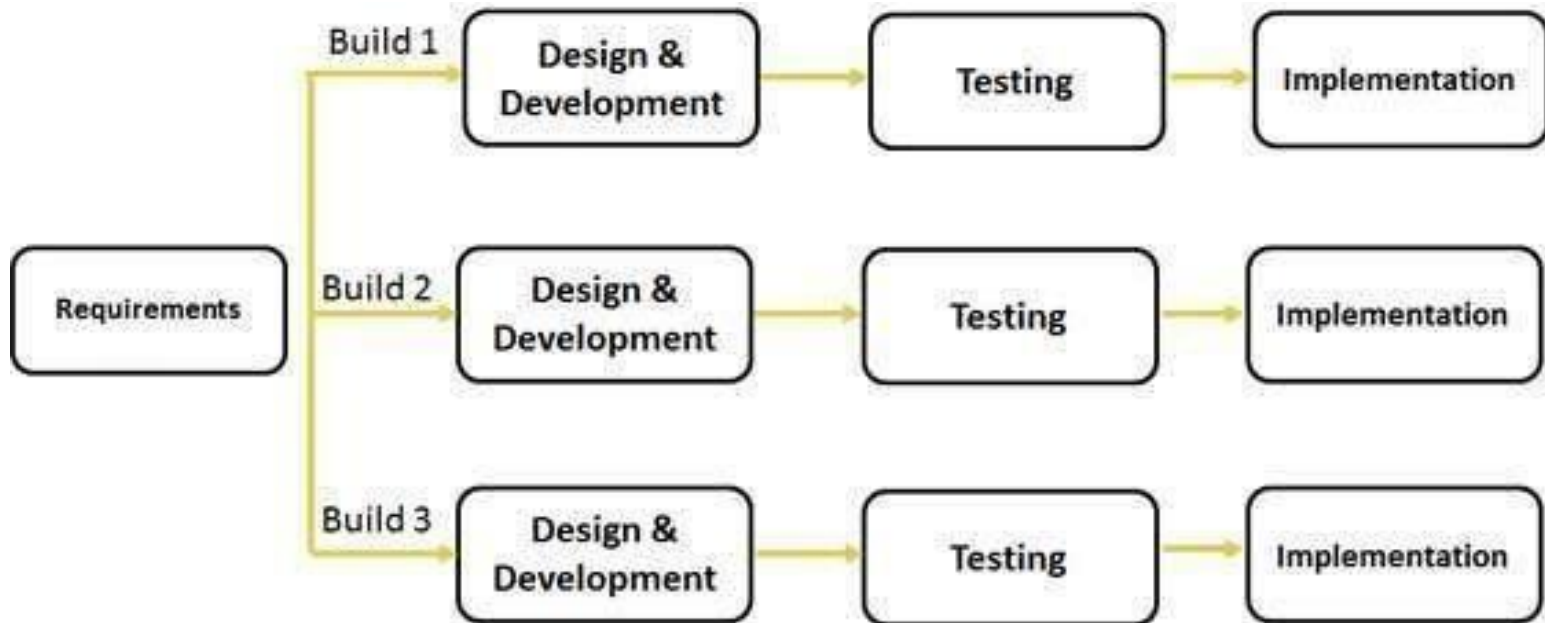
SDLC Models (Waterfall)

- The Phases are cascaded to each other in which progress is seen as flowing
- steadily downwards (like a waterfall) through the phases.
- The next phase is started only after the defined set of goals are achieved for previous phase and it is signed off, so the name "Waterfall Model". In this model phases do not overlap.



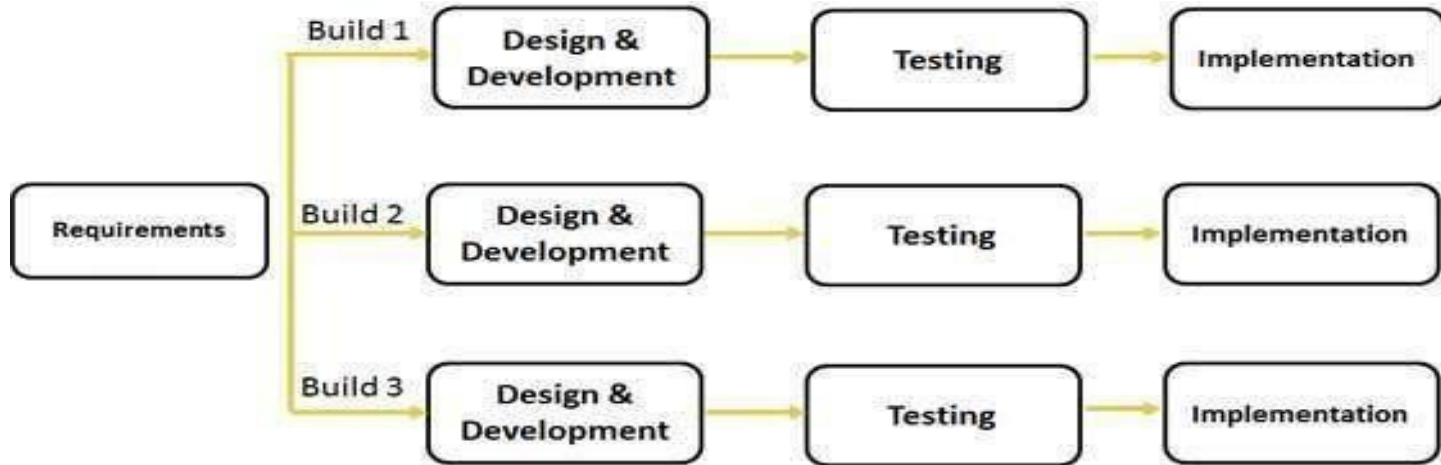
SDLC Models (Iterative)

- Requirements of the complete system are clearly defined and understood.
- Major requirements must be defined; however, some functionalities or requested enhancements may evolve with time.
- There is a time to the market constraint.

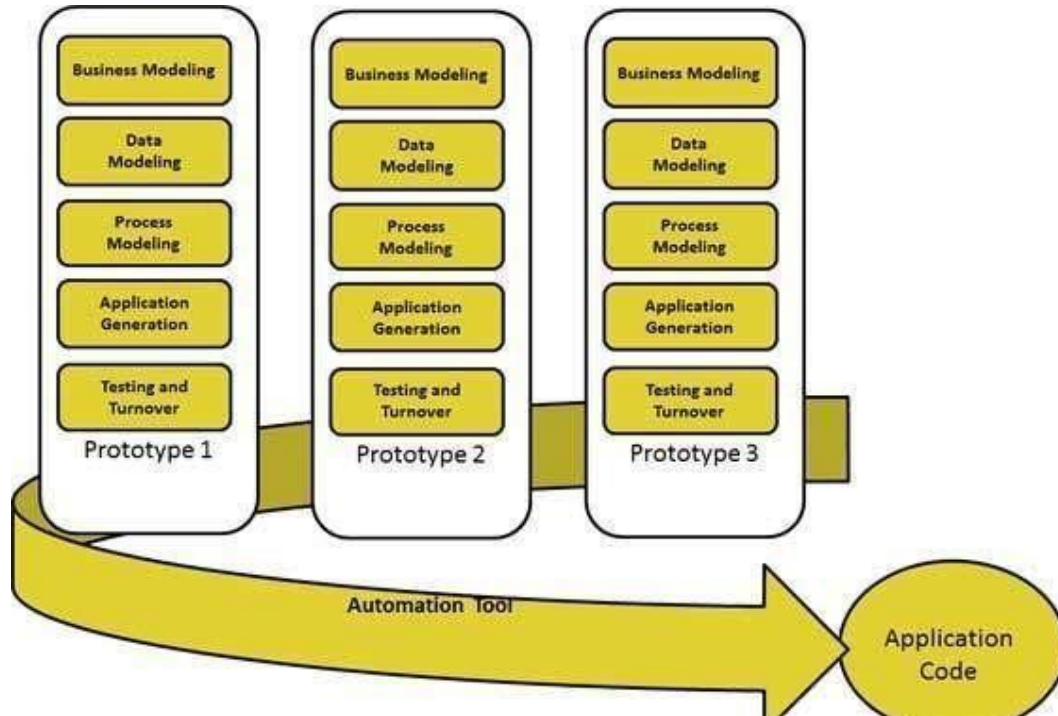


SDLC Models (Iterative)

- Resources with needed skill set are not available and are planned to be used on contract basis for specific iterations.
- There are some high risk features and goals which may change in the future.

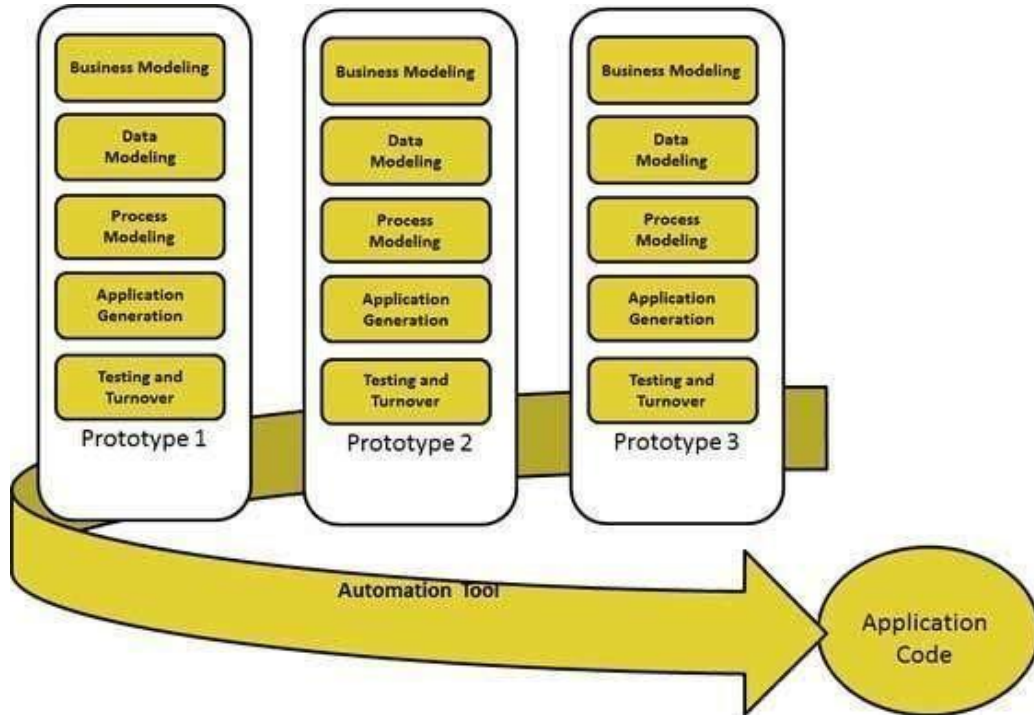


SDLC Models (RAD)



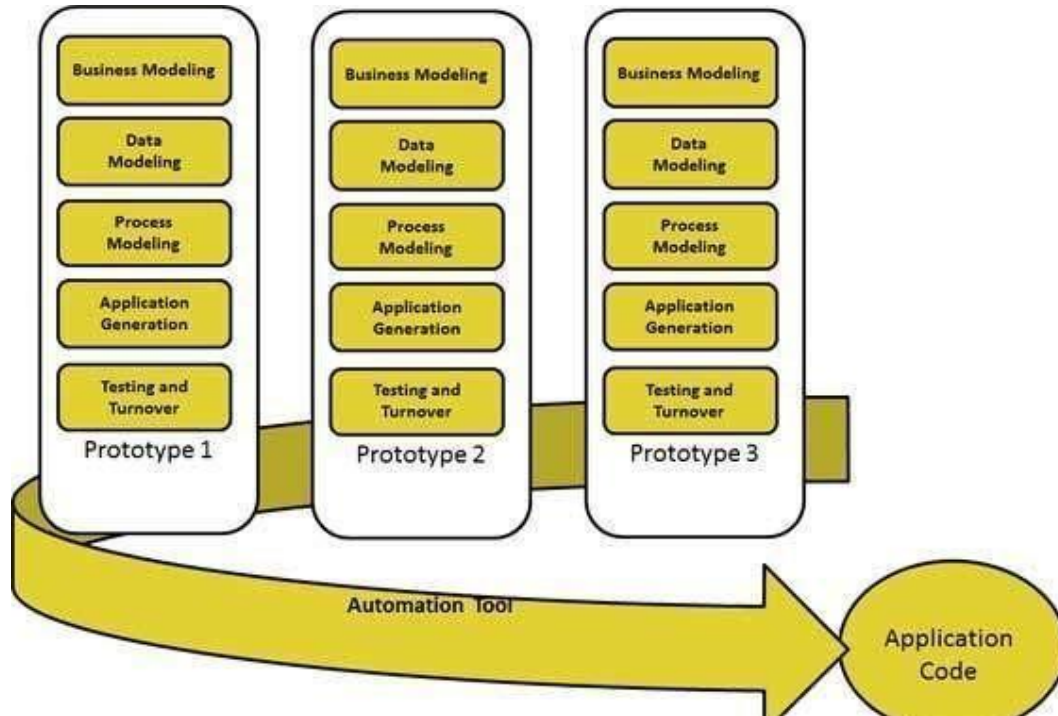
- RAD should be used only when a system can be modularized to be delivered in incremental manner.
- It should be used if there is high availability of designers for modeling.

SDLC Models (RAD)



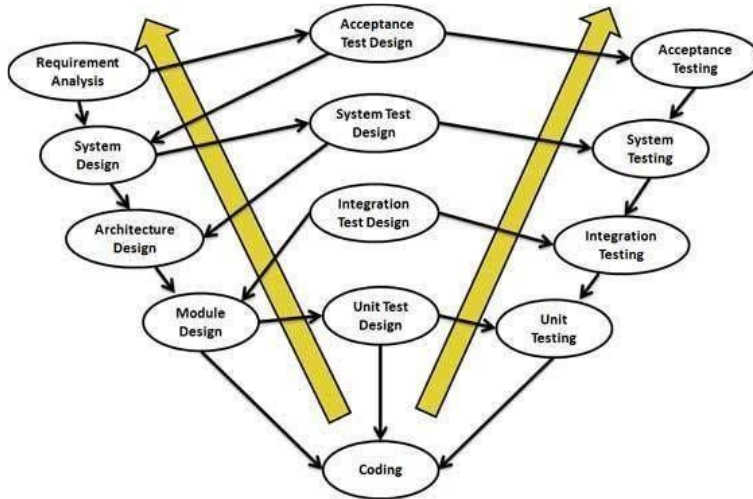
- It should be used only if the budget permits use of automated code generating tools.
- RAD SDLC model should be chosen only if domain experts are available with relevant business knowledge.

SDLC Models (RAD)



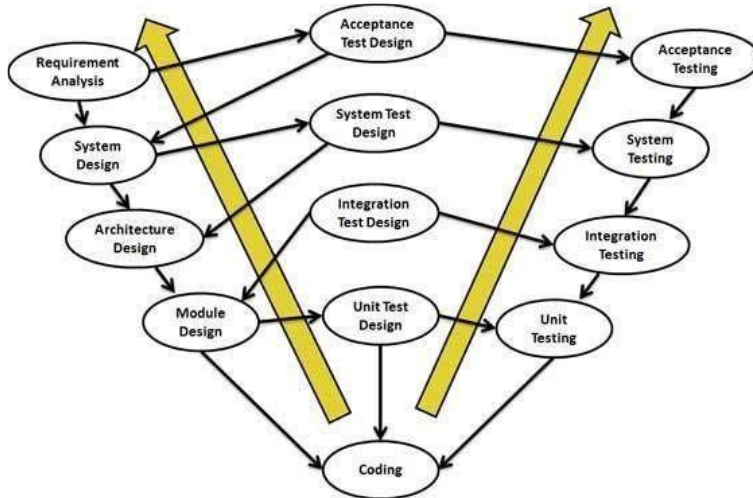
- Should be used where the requirements change during the course of the project and working prototypes are to be presented to customer in small iterations of 2-3 months.

SDLC Models (V)



- The V - model is SDLC model where execution of processes happens in a sequential manner in V-shape. It is also known as Verification and Validation model.
- V - Model is an extension of the waterfall model and is based on
- association of a testing phase for each corresponding development stage.

SDLC Models (V)



- This means that for every single phase in the development cycle there is a directly associated testing phase.
- This is a highly disciplined model and next phase starts only after completion of the previous phase.

SDLC Models (Prototype)

- The prototyping approach is used in the requirement gathering and in the analysis phase to capture the exact requirement of the proposed system.
- After the requirements are frozen, the remaining phases of the development process needs to be executed to complete the development of the software system.



Prototype Model

SDLC Models (Prototype)

- An e-commerce website, such as shopping site is an example where you can implement the prototyping approach. You can develop the prototype of the various web pages of the shopping site such as catalogue page, product order page etc., and present it to the customer for approved.

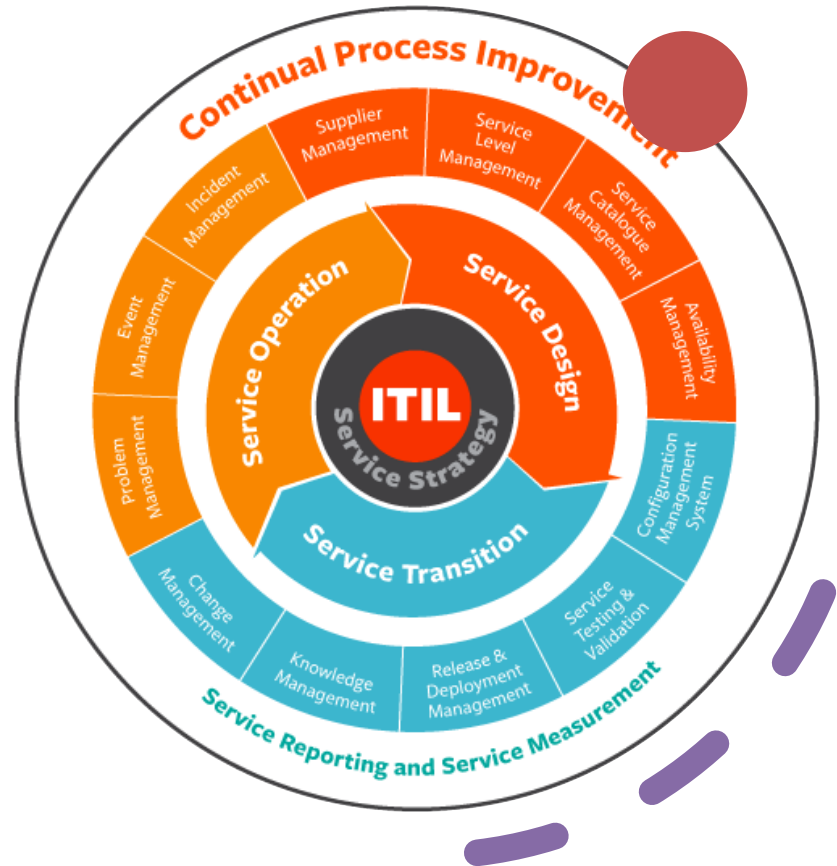
If the customer approves the prototype of the site, requirements are stated again and the design of the web site is initiated. If the customer does not approve the web site, the development team revisits the prototype and resubmits it to the customer for approval. This process continues until the prototype is approved.

Throwaway prototypes: Prototypes that are eventually discarded rather than becoming a part of the finally delivered software. Examples of throwaway prototypes include screen mock-ups and story boards.

Evolutionary Prototypes: prototypes that evolve into the final system through iterative incorporation of user feedback.

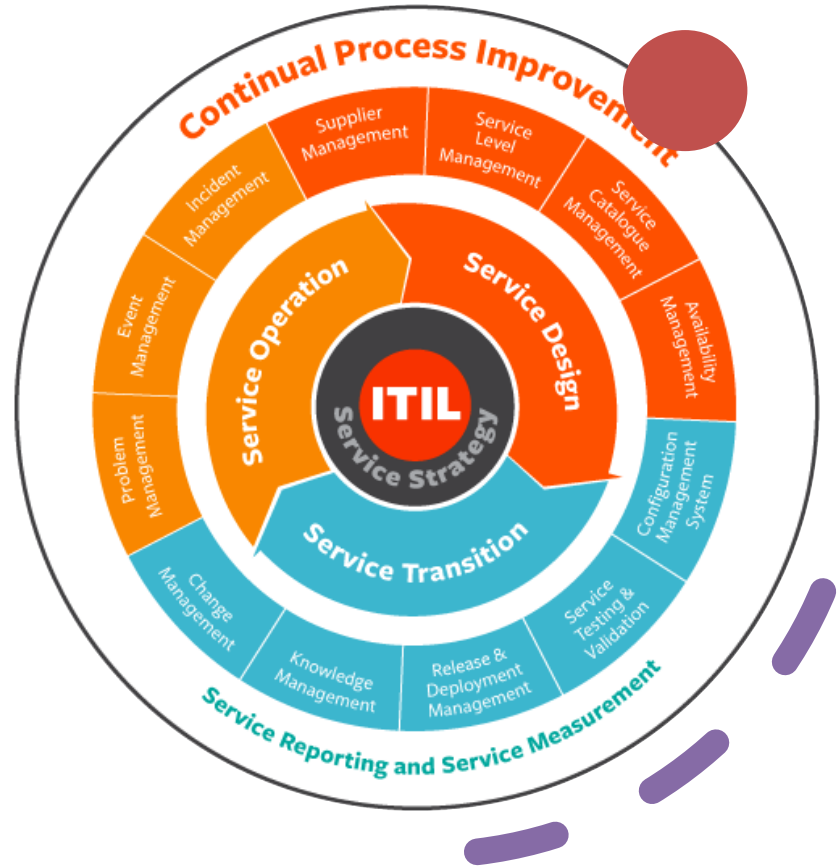
ITIL

- ITIL a.k.a. Information Technology Infrastructure Library helps individuals and organizations use IT to realize business change, transformation and growth.
- The ultimate goal of ITIL is to improve how IT delivers and supports valued business services.
- Every organization delivers a service or product. For every service or product, the ITIL framework helps manage delivery, industrialization, support, and consumerization from inception to retirement.



ITIL (Advantages)

- Stronger alignment between IT and the business.
- Improved service delivery and customer satisfaction.
- Reduced costs through improved use of resources.
- Greater visibility of IT costs and assets.
- Better management of business risk and service disruption or failure.
- More stable service environment to support constant business change.



The image features a white background with decorative geometric shapes in the corners. In the top-left corner, there are overlapping light blue and medium blue squares. In the top-right corner, there are overlapping light purple and medium purple squares. In the bottom-right corner, there are overlapping light blue and medium blue squares.

*Thank
you*